

Unit 1 - Quantum Mechanics

Quantum mechanics is a fundamental theory in physics that describes the behavior of matter and energy at the molecular, atomic, and subatomic level. It is based on the principles of wave-particle duality, uncertainty, and probability.

1. **Wave-Particle Duality:** This principle states that particles, such as electrons, can exhibit both wave-like and particle-like behavior. This is demonstrated by experiments such as the double-slit experiment, where electrons can create an interference pattern like waves, but also behave like particles when detected.
2. **Uncertainty Principle:** This principle, formulated by Werner Heisenberg, states that it is impossible to know both the position and momentum of a particle with absolute certainty. The more precisely one property is known, the less precisely the other can be known.
3. **Probability:** In quantum mechanics, the behavior of particles is described using probability waves, where the probability of finding a particle in a particular location is given by the square of the wave function. This means that the behavior of particles is inherently probabilistic, rather than deterministic.

These principles, along with others, form the basis of quantum mechanics and have been experimentally verified. Quantum mechanics has been incredibly successful in explaining a wide range of phenomena, from the behavior of subatomic particles to the properties of materials.

Inadequacy of Classical Mechanics

Classical mechanics is a branch of physics that explains the motion of celestial bodies such as planets, stars, and microscopic terrestrial bodies that move with non-relativistic speed. However, classical mechanics has its limitations and is inadequate in certain situations. Some of the inadequacies of classical mechanics are:

1. **It does not hold in the region of atomic dimensions.** Classical mechanics fails to explain the behavior of particles at the atomic and subatomic level. This led to the development of quantum mechanics, which provides a more accurate description of the behavior of particles at this scale.
2. **It could not explain the observed spectra of black body radiation.** Classical mechanics was unable to explain the observed spectra of black body radiation, which is the electromagnetic radiation emitted by a body in thermal equilibrium. This problem, known as the “ultraviolet catastrophe,” was eventually resolved by the development of quantum mechanics .

3. **It could not explain the photoelectric effect.** The photoelectric effect is the phenomenon where electrons are emitted from a material when light shines on it. Classical mechanics was unable to explain this effect, but it was eventually explained by the development of quantum mechanics.
4. **It could not explain the hydrogen emission (of light).** Classical mechanics was unable to explain the emission of light by hydrogen atoms. This problem was eventually resolved by the development of quantum mechanics.

These are some of the inadequacies of classical mechanics that led to the development of quantum mechanics, which provides a more accurate description of the behavior of particles at the atomic and subatomic level.

Planck's theory of black body radiation (qualitative)

- Planck's radiation law is a mathematical relationship formulated in 1900 by German physicist Max Planck to explain the spectral-energy distribution of radiation emitted by a blackbody.
- A blackbody is a hypothetical body that completely absorbs all radiant energy falling upon it, reaches some equilibrium temperature, and then reemits that energy as quickly as it absorbs it.
- Planck's law describes the spectral density of electromagnetic radiation emitted by a black body in thermal equilibrium at a given temperature T , when there is no net flow of matter or energy between the body and its environment.
- Using Planck's law, a black body's emitted radiation may be termed by its frequency variation. Planck's radiation, a form of thermal radiation, is another name for this type of radiation. The greater the temperature of the body, the more radiation of all wavelengths is emitted by it.
- The blackbody radiation problem was solved in 1900 by Max Planck. Planck used the same idea as the Rayleigh–Jeans model in the sense that he treated the electromagnetic waves between the walls inside the cavity classically, and assumed that the radiation is in equilibrium with the cavity walls.

Compton Effect

The Compton effect refers to the increase in the wavelength of photons (X-rays or gamma rays), due to their scattering by a charged particle (usually an electron). The effect has ended up being one of the foundations of quantum mechanics, which represents both wave and particle properties of radiation.

The American physicist Arthur Holly Compton clarified (in 1922 and published in 1923) the frequency increase by considering X-rays as made out of discrete pulses, or quanta, of electromagnetic energy.

Compton effect is the decrease in energy (increase in wavelength) of an X-ray or gamma ray photon, when it interacts with matter. Because of the change in photon energy, it is an inelastic scattering process. Inverse Compton scattering also exists, where the photon gains energy (decreasing in wavelength) upon interaction with matter.

Compton effect, also called Compton scattering, is a principal way in which radiant energy is absorbed in matter.

By classical theory, when an electromagnetic wave is scattered off atoms, the wavelength of the scattered radiation is expected to be the same as the wavelength of the incident radiation. However, the Compton effect is the term used for an unusual result observed when X-rays are scattered on some materials.

de-Broglie concept of matter waves

The de-Broglie concept of matter waves is a fundamental concept in quantum mechanics. It proposes that all matter exhibits wave-like properties and relates the observed wavelength of matter to its momentum.

- According to the de Broglie concept of matter waves, the matter has dual nature. It means when the matter is moving it shows the wave properties (like interference, diffraction etc.) are associated with it and when it is in the state of rest then it shows particle properties .
- Using the concept of the electron matter wave, de Broglie provided a rationale for the quantization of the electron's angular momentum in the hydrogen atom, which was postulated in Bohr's quantum theory.
- After Albert Einstein's photon theory became accepted, the question became whether this was true only for light or whether material objects also exhibited wave-like behavior.
- De Broglie, in his 1924 PhD thesis, proposed that just as light has both wave-like and particle-like properties, electrons also have wave-like properties.

This concept is important in understanding the behavior of particles at the quantum level and has significant implications in fields such as quantum mechanics and particle physics.

Davisson and Germer Experiment

The Davisson and Germer experiment was a 1923-27 experiment by Clinton Davisson and Lester Germer at Western Electric (later Bell Labs), in which electrons, scattered by the surface of a crystal of nickel metal, displayed a diffraction pattern.

The basic thought behind the experiment was that the waves reflected from two different atomic layers of a Ni crystal will have a fixed phase difference. After reflection, these waves will interfere either constructively or destructively, hence producing a diffraction pattern.

The Davisson and Germer experiment confirmed the de Broglie hypothesis that matter has wave-like behavior. This, in combination with the Compton effect discovered by Arthur Compton (who won the Nobel Prize for Physics in 1927), established the wave-particle duality hypothesis which was a fundamental step in quantum theory.

The Davisson and Germer Experiment established the wave nature of electrons and validated the de Broglie equation for the first time. De Broglie proposed the dual nature of the matter in 1924, but it wasn't until later that Davisson and Germer's experiment confirmed the findings.

Phase velocity and group velocity

Phase and group velocity are two important and related concepts in wave mechanics. They arise in quantum mechanics in the time development of the state function for the continuous case, i.e. wave packets.

Phase velocity

Phase velocity is the velocity at which the phase of a wave travels. It is described by the equation: $v_p = \omega/k$, where ω is the angular frequency of the wave and k is the wave number. The function $\omega(k)$, which gives ω as a function of k , is known as the dispersion relation.

Group velocity

Group velocity is equal to the energy at which the wave travels. If ω is directly proportional to k , then the group velocity is exactly equal to the phase velocity. A wave of any shape will travel undistorted at this velocity.

In a dispersive medium, the phase velocity depends on the frequency of the wave. In a non-dispersive medium, the phase velocity of a wave is independent of the frequency of the given wave.

Time-dependent and time-independent Schrodinger wave equations

The Schrodinger wave equation is a fundamental equation in quantum mechanics that describes how a quantum system changes over time. It is named after Erwin Schrodinger, who derived the equation in 1926.

There are two forms of the Schrodinger wave equation: the time-dependent and the time-independent Schrodinger wave equations.

Time-dependent Schrodinger wave equation

The time-dependent Schrodinger wave equation describes the evolution of the wave function of a quantum system over time. It is given by the equation:

$$i\hbar \frac{\partial \psi}{\partial t} = H \psi$$

where i is the imaginary unit, $\hbar$ is the reduced Planck constant, ψ is the wave function of the system, t is time, and H is the Hamiltonian operator, which represents the total energy of the system.

Time-independent Schrodinger wave equation

The time-independent Schrodinger wave equation describes the stationary states of a quantum system. It is obtained by assuming that the wave function can be separated into a time-dependent part and a time-independent part, and then applying the time-dependent Schrodinger wave equation. The resulting equation is:

$$H \psi = E \psi$$

where E is the total energy of the system.

The time-independent Schrodinger wave equation is often used to solve for the stationary states of a quantum system, which are the states with definite energy. These states are important because they form a complete set of states that can be used to describe any state of the system.

In summary, the Schrodinger wave equation is a fundamental equation in quantum mechanics that describes how a quantum system changes over time. There are two forms of the Schrodinger wave equation: the time-dependent and the time-independent Schrodinger wave equations. The time-dependent Schrodinger wave equation describes the evolution of the wave function of a quantum system over time, while the time-independent Schrodinger wave equation describes the stationary states of a quantum system. These equations are important tools for understanding and predicting the behavior of quantum systems.

Physical interpretation of wave function

- The wave function, at a particular time, contains all the information that anybody at that time can have about the particle .
- The wave function itself has no physical interpretation. It is not measurable .
- However, the square of the absolute value of the wave function has a physical interpretation .

- The standard assumption is that the wave function of an electron is a probability amplitude, and its modulus square gives the probability density of finding the electron in a certain location at a given instant .

Particle in a One-Dimensional Box

The particle in a one-dimensional box is a fundamental problem in quantum mechanics. It is used to illustrate the basic principles of quantum mechanics and is often used as a starting point for more complex problems.

The problem involves a particle that is confined to a one-dimensional box of length L . The potential energy of the particle is zero inside the box and infinite outside the box. This means that the particle is free to move inside the box, but cannot escape from it.

The wave function of the particle must satisfy the Schrödinger equation, which can be written as:

$$-\hbar^2/2m * d^2(\psi)/dx^2 + V(x) \psi = E \psi$$

where $\hbar$ is the reduced Planck constant, m is the mass of the particle, $V(x)$ is the potential energy, and E is the total energy of the particle.

Since the potential energy is zero inside the box, the Schrödinger equation can be simplified to:

$$-\hbar^2/2m * d^2(\psi)/dx^2 = E \psi$$

This is a second-order differential equation, which has the general solution:

$$\psi(x) = A * \sin(kx) + B * \cos(kx)$$

where A and B are constants and k is a constant related to the energy of the particle.

The boundary conditions of the problem require that the wave function must be zero at the edges of the box, i.e., $\psi(0) = \psi(L) = 0$. This means that B must be zero and that k must satisfy the condition:

$$kL = n$$

where n is a positive integer.

The energy of the particle is given by:

$$E = \hbar^2 k^2 / 2m = n^2 \pi^2 \hbar^2 / 2mL^2$$

This shows that the energy of the particle is quantized, i.e., it can only take on certain discrete values. The lowest energy state, called the ground state, corresponds to $n = 1$. The energy of the ground state is:

$$E_1 = \pi^2 \hbar^2 / 2mL^2$$

The next energy level, called the first excited state, corresponds to $n = 2$. The energy of the first excited state is:

$$E = 4 \pi^2 \hbar^2 / 2mL^2$$

In general, the energy of the n th excited state is:

$$E = n^2 \pi^2 \hbar^2 / 2mL^2$$

The wave function of the particle in the n th energy state is given by:

$$\psi(x) = \sqrt{2/L} * \sin(n\pi x/L)$$

Unit 2 - Electromagnetic Field Theory

1. Electromagnetic Field Theory is the study of electromagnetic fields and their interactions with charged particles.
2. The fundamental equations governing electromagnetic fields are Maxwell's equations, which describe the relationship between electric and magnetic fields, and their sources, electric charges and currents.
3. Electromagnetic fields can be divided into two categories: static fields and time-varying fields. Static fields do not change with time, while time-varying fields do.
4. The electric field is a vector field that describes the force exerted on a charged particle by other charged particles. The electric field is defined as the force per unit charge.
5. The magnetic field is a vector field that describes the force exerted on a moving charged particle by other moving charged particles. The magnetic field is defined as the force per unit charge per unit velocity.
6. Electromagnetic waves are transverse waves that consist of oscillating electric and magnetic fields. They can travel through a vacuum and do not require a medium to propagate.
7. The speed of electromagnetic waves in a vacuum is the speed of light, which is approximately 299,792,458 meters per second.
8. Electromagnetic waves can have different frequencies and wavelengths, and the electromagnetic spectrum includes radio waves, microwaves, infrared radiation, visible light, ultraviolet radiation, X-rays, and gamma rays.
9. Electromagnetic fields have many practical applications, including in communication, medicine, and energy generation.

Basic concept of Stoke's theorem and Divergence theorem for the notes of the Unit 2 - Electromagnetic Field Theory in the subject of ENGINEERING PHYSICS

Stoke's Theorem: - Stoke's theorem is a statement about the integration of differential forms on manifolds. - It relates the integral of a differential form over the boundary of some orientable manifold to the integral of its exterior derivative over the whole manifold. - In simple terms, it states that the integral

of a curl of a vector field over a surface is equal to the line integral of the vector field around the boundary of the surface.

Divergence Theorem: - The divergence theorem, also known as Gauss's theorem, is a result that relates the flow of a vector field through a closed surface to the behavior of the vector field inside the surface. - It states that the integral of the divergence of a vector field over a volume is equal to the flux of the vector field through the boundary of the volume. - This theorem is widely used in physics and engineering to calculate the flux of a vector field through a closed surface.

Both Stoke's theorem and the divergence theorem are important tools in the study of electromagnetic fields. They allow for the calculation of integrals over surfaces and volumes, which is essential in the analysis of electric and magnetic fields. These theorems are covered in Unit 2 of the subject of Engineering Physics, under the topic of Electromagnetic Field Theory.

Basic Laws of Electricity and Magnetism

Unit 2 - Electromagnetic Field Theory in the subject of ENGINEERING PHYSICS

1. **Coulomb's Law:** This law states that the force between two point charges is directly proportional to the product of the magnitudes of the charges and inversely proportional to the square of the distance between them. The force acts along the line joining the two charges.
2. **Ohm's Law:** This law states that the current flowing through a conductor is directly proportional to the potential difference applied across its ends, provided the physical conditions, such as temperature, remain constant.
3. **Ampere's Law:** This law states that the magnetic field produced by an electric current is proportional to the current and encircles the current-carrying conductor.
4. **Faraday's Law of Electromagnetic Induction:** This law states that a change in the magnetic field within a closed loop of wire induces an electromotive force (EMF) in the wire. This induced EMF is proportional to the rate of change of the magnetic flux through the loop.
5. **Lenz's Law:** This law states that the direction of the induced EMF and hence the induced current in a closed circuit is such as to oppose the change in magnetic flux that produces it.
6. **Biot-Savart Law:** This law states that the magnetic field produced by a current-carrying element is directly proportional to the current and inversely proportional to the square of the distance from the element.
7. **Gauss's Law for Magnetism:** This law states that the net magnetic flux through any closed surface is zero. This implies that magnetic monopoles

do not exist.

These are some of the basic laws of electricity and magnetism that are covered in Unit 2 - Electromagnetic Field Theory in the subject of ENGINEERING PHYSICS. These laws form the foundation for the study of electromagnetism and are essential for understanding the behavior of electric and magnetic fields.

Continuity Equation for Current Density

The continuity equation for current density is an important concept in the study of Electromagnetic Field Theory, which is a part of the subject of Engineering Physics. Here are some key points to remember:

1. The continuity equation for current density is derived from the principle of conservation of charge, which states that the total electric charge in an isolated system remains constant.
2. The equation states that the rate of change of electric charge in a given volume is equal to the net current flowing into the volume.
3. Mathematically, the continuity equation for current density can be expressed as: $\nabla \cdot \mathbf{J} + \frac{\partial \rho}{\partial t} = 0$, where $\mathbf{J}$ is the current density vector, ρ is the charge density, and t is time.
4. The equation can be interpreted as follows: the divergence of the current density vector at any point is equal to the negative rate of change of charge density at that point.
5. The continuity equation for current density is an important tool for analyzing the behavior of electric currents in various situations, including the flow of current in circuits and the propagation of electromagnetic waves.

Displacement current

- Displacement current is a phenomenon in electromagnetism that is analogous to an ordinary electric current. It is posited to explain magnetic fields that are produced by changing electric fields.
- Displacement current density is the quantity $\partial \mathbf{D} / \partial t$ appearing in Maxwell's equations that is defined in terms of the rate of change of $\mathbf{D}$, the electric displacement field.
- Displacement current density has the same units as electric current density, and it is a source of the magnetic field just as actual current is.
- A traveling and varying magnetic field is related to a synchronous change in the electric field that results in displacement current.
- The displacement current plays a major role during the propagation of electromagnetic radiations like light waves and radio waves through space.

Maxwell Equations in Integral and Differential Form

Maxwell's equations are a set of four partial differential equations that describe the behavior of the electric and magnetic fields. These equations were first presented by James Clerk Maxwell in the 19th century. They are fundamental to the study of electromagnetic field theory and are widely used in electrical engineering, physics, and many other fields.

The four Maxwell equations in integral form are:

1. Gauss's Law: This law states that the electric flux through any closed surface is equal to the charge enclosed within the surface divided by the permittivity of free space. Mathematically, it is expressed as:

$$\oint \vec{E} \cdot d\vec{A} = \frac{Q}{\epsilon_0}$$

2. Gauss's Law for Magnetism: This law states that the magnetic flux through any closed surface is zero. Mathematically, it is expressed as:

$$\oint \vec{B} \cdot d\vec{A} = 0$$

3. Faraday's Law of Induction: This law states that the electromotive force (EMF) induced in a closed loop is equal to the negative of the time rate of change of the magnetic flux through the loop. Mathematically, it is expressed as:

$$\oint \vec{E} \cdot d\vec{l} = -\frac{d\Phi_B}{dt}$$

4. Ampere's Law with Maxwell's Correction: This law states that the magnetic field around a closed loop is equal to the sum of the current flowing through the loop and the displacement current, which is proportional to the time rate of change of the electric flux through the loop. Mathematically, it is expressed as:

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 \left(I + \epsilon_0 \frac{d\Phi_E}{dt} \right)$$

The four Maxwell equations in differential form are:

1. Gauss's Law: This law states that the divergence of the electric field is equal to the charge density divided by the permittivity of free space. Mathematically, it is expressed as:

$$\nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0}$$

2. Gauss's Law for Magnetism: This law states that the divergence of the magnetic field is zero. Mathematically, it is expressed as:

$$\nabla \cdot \vec{B} = 0$$

3. Faraday's Law of Induction: This law states that the curl of the electric field is equal to the negative of the time rate of change of the magnetic field. Mathematically, it is expressed as:

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

4. Ampere's Law with Maxwell's Correction: This law states that the curl of the magnetic field is equal to the current density plus the displacement current density, which is proportional to the time rate of change of the electric field. Mathematically, it is expressed as:

$$\nabla \times \vec{B} = \mu_0 \left(\vec{J} + \epsilon_0 \frac{\partial \vec{E}}{\partial t} \right)$$

These equations can be used to describe the behavior of electromagnetic fields in various situations, including the propagation of electromagnetic waves, the interaction of charged particles with electromagnetic fields, and the generation of electric and magnetic fields by charges and currents. They are essential for understanding the principles of electromagnetic field theory and are widely used in the study of electrical engineering and physics.

Maxwell equations in vacuum and in conducting medium

Maxwell's equations are a set of four partial differential equations that describe the behavior of the electric and magnetic fields. These equations were first presented by James Clerk Maxwell in the 19th century. They are fundamental to the study of electromagnetic field theory and are widely used in physics and engineering.

In a vacuum, the current density, $\vec{J}$, is equal to 0. Maxwell added a new term called displacement current into Ampere's law, forming Maxwell's fourth equation. The electromagnetic wave equation derives from Maxwell's equations. At every point in space and at every time, the field vectors satisfy the Maxwell equations.

Maxwell's equations may be combined to demonstrate how fluctuations in electromagnetic fields (waves) propagate at a constant speed in vacuum, c (299792458 m/s). Known as electromagnetic radiation, these waves occur at

various wavelengths to produce a spectrum of radiation from radio waves to gamma rays.

In a conducting medium, Maxwell equations can be modified by including to $\mathbf{U}$ and $\mathbf{j}$ & charges and currents inside atoms. In this case fields (so-called microscopic fields) are changing on microscopic scales comparable with atomic size and number of sources is too large.

Poynting Vector and Poynting Theorem

- The Poynting vector represents the directional energy flux or power flow of an electromagnetic field. The SI unit of the Poynting vector is the watt per square metre (W/m^2) .
- The Poynting vector is used throughout electromagnetics in conjunction with Poynting's theorem, the continuity equation expressing conservation of electromagnetic energy, to calculate the power flow in electric and magnetic fields .
- Poynting's theorem is a statement of conservation of energy for electromagnetic fields developed by British physicist John Henry Poynting. It states that in a given volume, the stored energy changes at a rate given by the work done on the charges within the volume, minus the rate at which energy leaves the volume .
- Poynting's theorem states that the cross product of electric field vector, $\mathbf{E}$ and magnetic field vector, $\mathbf{H}$ at any point is a measure of the rate of flow of electromagnetic energy per unit area at that point, that is $\mathbf{P} = \mathbf{E} \times \mathbf{H}$. Here $\mathbf{P}$ is the Poynting vector and it is named after its discoverer, J.H. Poynting .
- Poynting's vector is a measure of the momentum density carried by the electromagnetic field. Momentum density in the field is given by $\mathbf{g} = \mathbf{S}/c^2$.

Plane Electromagnetic Waves in Vacuum and their Transverse Nature

Unit 2 - Electromagnetic Field Theory in the subject of ENGINEERING PHYSICS

1. An electromagnetic wave is a transverse wave that consists of oscillating electric and magnetic fields that propagate through space.
2. In a vacuum, electromagnetic waves travel at the speed of light, which is approximately 299,792,458 meters per second.
3. The electric and magnetic fields of an electromagnetic wave are perpendicular to each other and to the direction of propagation of the wave.
4. This means that the electric and magnetic fields oscillate in planes that are perpendicular to the direction in which the wave is traveling.

5. The transverse nature of electromagnetic waves can be demonstrated by the polarization of light.
6. Polarization refers to the orientation of the electric field of an electromagnetic wave.
7. When light passes through a polarizing filter, only the component of the electric field that is parallel to the filter is transmitted, while the component that is perpendicular to the filter is absorbed.
8. This results in the light being polarized in the direction of the filter.
9. The transverse nature of electromagnetic waves has important implications for the behavior of these waves when they interact with matter.
10. For example, the reflection and refraction of light can be understood in terms of the interaction of the electric and magnetic fields of the electromagnetic wave with the electrons in the atoms of the material.

Relation between electric and magnetic fields of an electromagnetic wave

An electromagnetic wave is a type of wave that consists of oscillating electric and magnetic fields that travel through space. The electric and magnetic fields of an electromagnetic wave are perpendicular to each other and to the direction of the wave's propagation.

The relationship between the electric and magnetic fields of an electromagnetic wave can be described by Maxwell's equations. These equations show that a changing electric field generates a magnetic field, and a changing magnetic field generates an electric field. This means that the electric and magnetic fields of an electromagnetic wave are interdependent and cannot exist without each other.

In an electromagnetic wave, the magnitude of the electric field is directly proportional to the magnitude of the magnetic field. The direction of the electric field is perpendicular to the direction of the magnetic field, and both fields are perpendicular to the direction of the wave's propagation.

In summary, the electric and magnetic fields of an electromagnetic wave are interdependent and are related by Maxwell's equations. The magnitude of the electric field is directly proportional to the magnitude of the magnetic field, and the direction of the electric field is perpendicular to the direction of the magnetic field. Both fields are perpendicular to the direction of the wave's propagation. This relationship is an important concept in the study of electromagnetic field theory.

Plane Electromagnetic Waves in Conducting Medium

Unit 2 - Electromagnetic Field Theory in the subject of ENGINEERING PHYSICS

1. A conducting medium is a material that allows the flow of electric charge.

2. When an electromagnetic wave travels through a conducting medium, the electric field of the wave causes the free electrons in the medium to oscillate.
3. This oscillation of electrons produces an electric current, which in turn generates a magnetic field.
4. The magnetic field then interacts with the electric field of the wave, causing the wave to be attenuated or weakened as it travels through the medium.
5. The attenuation of the wave is determined by the conductivity of the medium, with higher conductivity leading to greater attenuation.
6. The speed of the wave is also affected by the conductivity of the medium, with the wave traveling slower in a more conductive medium.
7. The interaction between the electromagnetic wave and the conducting medium can be described by the wave equation, which takes into account the conductivity, permittivity, and permeability of the medium.
8. Solutions to the wave equation can be used to determine the behavior of electromagnetic waves in conducting media, including reflection, refraction, and transmission.

Skin Depth

Skin depth is a measure of the distance an electromagnetic wave can penetrate into a conductor before it is attenuated to $1/e$ (about 37%) of its original amplitude. It is a key concept in electromagnetic field theory and is used to analyze the behavior of electromagnetic waves in conductive materials.

1. Skin depth is defined as the distance over which the amplitude of an electromagnetic wave decreases to $1/e$ of its original value.
2. The skin depth is dependent on the frequency of the electromagnetic wave, the conductivity of the material, and the permeability of the material.
3. The skin depth decreases as the frequency of the electromagnetic wave increases, meaning that higher frequency waves penetrate less into a conductor.
4. The skin depth also decreases as the conductivity of the material increases, meaning that more conductive materials have a smaller skin depth.
5. The skin depth is also affected by the permeability of the material, with materials with higher permeability having a smaller skin depth.

In summary, skin depth is an important concept in electromagnetic field theory that describes the distance an electromagnetic wave can penetrate into a conductor before being significantly attenuated. It is dependent on the frequency of the wave, the conductivity of the material, and the permeability of the material. Understanding skin depth is crucial for the analysis of electromagnetic wave behavior in conductive materials.

Unit 3 - Wave Optics

Wave optics, also known as physical optics, is the branch of optics that studies the behavior of light as a wave. This includes phenomena such as interference, diffraction, and polarization.

1. **Interference:** Interference is the phenomenon in which two or more waves superimpose to form a resultant wave of greater, lower, or the same amplitude. Constructive interference occurs when the phase difference between the waves is a multiple of 2π , whereas destructive interference occurs when the phase difference is an odd multiple of π .
2. **Diffraction:** Diffraction is the bending of waves around obstacles or the spreading of waves when they pass through an aperture. The amount of diffraction depends on the wavelength of the wave and the size of the obstacle or aperture.
3. **Polarization:** Polarization is the property of electromagnetic waves, such as light, that describes the direction of the electric field. Unpolarized light consists of waves with electric fields in all directions, whereas polarized light consists of waves with electric fields in a specific direction.

Coherent sources for the notes of the Unit 3 - Wave Optics in the subject of ENGINEERING PHYSICS

- Coherent sources are sources of light that emit waves of the same frequency and constant phase difference.
- The waves from coherent sources interfere with each other to produce a stable interference pattern.
- Coherent sources can be obtained by dividing the light from a single source into two or more parts, using devices such as beam splitters or mirrors.
- Lasers are an example of a source of coherent light.
- Coherence is important in applications such as holography, interferometry, and optical communication.

Interference in Uniform and Wedge Shaped Thin Films

Interference is a phenomenon in which two or more waves superimpose to form a resultant wave of greater, lower, or the same amplitude. Interference usually refers to the interaction of waves that are correlated or coherent with each other, either because they come from the same source or because they have the same or nearly the same frequency.

In the context of uniform and wedge shaped thin films, interference occurs when light waves reflect off the top and bottom surfaces of a thin film. The waves that reflect off the bottom surface travel a slightly longer distance than the waves that reflect off the top surface, causing a phase difference between the two waves. This phase difference can result in constructive or destructive interference, depending on the thickness of the film and the wavelength of the light.

In a uniform thin film, the thickness of the film is constant, resulting in a constant phase difference between the two waves. This can result in a series of bright and dark bands, known as interference fringes, when the film is viewed under monochromatic light.

In a wedge shaped thin film, the thickness of the film varies, resulting in a varying phase difference between the two waves. This can result in a more complex interference pattern, with the fringes becoming closer together as the thickness of the film decreases.

Interference in thin films has many practical applications, including the creation of anti-reflective coatings, the measurement of film thickness, and the production of optical filters.

Some key points to remember about interference in uniform and wedge shaped thin films are: - Interference occurs when light waves reflect off the top and bottom surfaces of a thin film. - The phase difference between the two waves can result in constructive or destructive interference. - In a uniform thin film, the thickness of the film is constant, resulting in a constant phase difference between the two waves. - In a wedge shaped thin film, the thickness of the film varies, resulting in a varying phase difference between the two waves. - Interference in thin films has many practical applications, including the creation of anti-reflective coatings, the measurement of film thickness, and the production of optical filters.

Necessity of extended sources for the notes of the Unit 3 - Wave Optics in the subject of ENGINEERING PHYSICS

1. Extended sources provide a more comprehensive understanding of the subject matter.
2. They offer a wider range of perspectives and approaches to the topic.
3. They can provide additional examples and explanations to supplement the main course material.
4. They can help students to develop a deeper understanding of the concepts and their applications.
5. They can provide additional practice problems and exercises to reinforce learning.
6. They can help students to prepare for exams by providing additional review material.
7. They can provide a broader context for the subject matter, helping students to see the connections between different topics.
8. They can help students to stay up-to-date with the latest developments and research in the field.

Newton's Rings and its applications

Newton's rings is a phenomenon in which an interference pattern is created by the reflection of light between two surfaces - a spherical surface and an adjacent flat surface. It is named after Sir Isaac Newton, who first studied the effect in the 17th century.

The phenomenon is observed due to the interference of light waves reflected from the top and bottom surfaces of a thin film of air between the two surfaces. When viewed with monochromatic light, the pattern appears as a series of concentric, alternating bright and dark rings centered at the point of contact between the two surfaces.

Applications of Newton's Rings include: 1. Determining the wavelength of light: The diameter of the rings in the Newton's Rings pattern is directly proportional to the square root of the wavelength of the light used. By measuring the diameter of the rings, the wavelength of the light can be determined. 2. Measuring the refractive index of a transparent material: The refractive index of a transparent material can be determined by placing a thin film of the material between the two surfaces and observing the change in the diameter of the rings. 3. Testing the quality of optical surfaces: Newton's Rings can be used to test the quality of optical surfaces, such as lenses and mirrors. Any deviation from a perfect surface will cause a distortion in the pattern of the rings.

Introduction to Diffraction

Diffraction is the phenomenon of bending of light around the corners of an obstacle or aperture into the region of geometrical shadow of the obstacle. It is a characteristic property of waves, including light waves, sound waves, and water waves. Diffraction is observed when the size of the obstacle or aperture is comparable to the wavelength of the wave.

Some key points to remember about diffraction are: - Diffraction is a result of the wave nature of light. - The amount of diffraction depends on the size of the obstacle or aperture relative to the wavelength of the wave. - The smaller the obstacle or aperture, the greater the diffraction. - Diffraction is more pronounced for longer wavelengths.

In the context of wave optics, diffraction is an important phenomenon to understand as it helps to explain the behavior of light as it interacts with various optical elements. It is also essential for understanding the limitations of optical instruments, such as microscopes and telescopes.

Fraunhofer Diffraction at Single Slit and Double Slit

Fraunhofer diffraction is a type of diffraction that occurs when a wave passes through an aperture or slit and the distance between the aperture and the screen is much larger than the size of the aperture. This type of diffraction is also known as far-field diffraction.

Single Slit Diffraction

When a monochromatic light wave passes through a single slit, the wave spreads out and forms a diffraction pattern on a screen placed at a distance from the slit. The diffraction pattern consists of a central bright fringe, known as the central maximum, and a series of alternating bright and dark fringes on either side of the central maximum.

The width of the central maximum is inversely proportional to the width of the slit. As the width of the slit decreases, the width of the central maximum increases. The intensity of the central maximum is also affected by the width of the slit. As the width of the slit decreases, the intensity of the central maximum decreases.

The position of the bright fringes in the diffraction pattern can be calculated using the formula:

$$d \sin(\theta) = m \lambda$$

where d is the width of the slit, θ is the angle between the central maximum and the bright fringe, m is the order of the fringe, and λ is the wavelength of the light.

Double Slit Diffraction

When a monochromatic light wave passes through two slits, the waves from each slit interfere with each other to form an interference pattern on a screen placed at a distance from the slits. The interference pattern consists of a series of bright and dark fringes.

The position of the bright fringes in the interference pattern can be calculated using the formula:

$$d \sin(\theta) = m \lambda$$

where d is the distance between the two slits, θ is the angle between the central maximum and the bright fringe, m is the order of the fringe, and λ is the wavelength of the light.

The width of the bright fringes in the interference pattern is inversely proportional to the distance between the two slits. As the distance between the two slits decreases, the width of the bright fringes increases.

The intensity of the bright fringes in the interference pattern is also affected by the width of the slits. As the width of the slits decreases, the intensity of the bright fringes decreases.

In summary, Fraunhofer diffraction at a single slit and double slit is a phenomenon that occurs when a wave passes through an aperture or slits and forms a diffraction or interference pattern on a screen placed at a distance from the aperture or slits. The position, width, and intensity of the fringes in the pattern are affected by the width of the slit or the distance between the two slits, and the wavelength of the light. This is an important concept in the study of wave optics in engineering physics.

Absent Spectra

- Absent spectra, also known as absorption spectra, are produced when light passes through a medium and certain wavelengths are absorbed by the medium.
- This results in dark lines or bands in the spectrum, where the absorbed wavelengths would have been.
- Absent spectra are the opposite of emission spectra, where light is emitted by a medium at specific wavelengths, resulting in bright lines or bands in the spectrum.
- Absent spectra can be used to identify the composition of a medium, as different elements and compounds absorb light at specific wavelengths.
- In the context of wave optics, absent spectra can be observed when white light passes through a prism and is dispersed into its component colors. If the light passes through a medium before reaching the prism, certain wavelengths may be absorbed, resulting in an absent spectrum.
- Absent spectra are important in fields such as astronomy, where they are used to determine the composition of stars and other celestial objects.

Diffraction Grating

A diffraction grating is an optical component with a periodic structure that splits and diffracts light into several beams traveling in different directions. The directions of these beams depend on the spacing of the grating and the wavelength of the light so that the grating acts as the dispersive element.

Diffraction gratings are used in spectrometers, monochromators, and other optical instruments to separate light into its component wavelengths. They can also be used to produce diffraction patterns, which can be used to measure the properties of light and the materials it interacts with.

The basic principle behind the diffraction grating is the interference of light waves. When light waves encounter a periodic structure, such as the lines on a diffraction grating, they interfere with each other to produce a pattern of bright

and dark regions. The spacing of these regions depends on the wavelength of the light and the spacing of the grating lines.

Diffraction gratings can be made by ruling lines on a reflective surface or by using a holographic process to create a pattern of lines on a transparent material. The spacing of the lines determines the dispersion of the grating, with finer spacings producing greater dispersion.

In summary, a diffraction grating is an optical component that separates light into its component wavelengths by exploiting the interference of light waves. It is used in a variety of optical instruments to analyze the properties of light and the materials it interacts with. The spacing of the grating lines determines the dispersion of the grating, with finer spacings producing greater dispersion.

Spectra with Grating

1. A diffraction grating is an optical component with a periodic structure that splits and diffracts light into several beams traveling in different directions. The directions of these beams depend on the spacing of the grating and the wavelength of the light so that the grating acts as the dispersive element.
2. The diffraction pattern formed with a grating is known as a grating spectrum. The diffraction grating spectrometer has a superiority $S = \tan i$, where i is the angular length of the slit and i is the angle of incidence onto the grating.
3. Diffraction gratings are commonly used for spectroscopic dispersion and analysis of light. What makes them particularly useful is the fact that they form a sharper pattern than double slits do. That is, their bright fringes are narrower and brighter while their dark regions are darker.
4. If the number of lines in the grating is very large, then the maxima are sharp and bright. This will be an extremely useful feature.

Dispersive Power

Dispersive power is a measure of the ability of a medium to separate light of different wavelengths by refraction. It is defined as the ratio of the difference in the angles of deviation for two rays of light of different wavelengths to the difference in the angles of minimum deviation for the same two rays.

In mathematical terms, the dispersive power (ω) of a medium is given by the formula:

$$\omega = (\delta_1 - \delta_2) / (D_{m1} - D_{m2})$$

where δ_1 and δ_2 are the angles of deviation for two rays of light of different wavelengths, and D_{m1} and D_{m2} are the angles of minimum deviation for the same two rays.

The dispersive power of a medium is an important property in the field of optics, as it determines the degree to which a prism or lens will separate light into its

constituent colors. A medium with a high dispersive power will produce a more pronounced separation of colors than a medium with a low dispersive power.

In the context of wave optics, the dispersive power of a medium is related to its refractive index and the wavelength of light. The refractive index of a medium generally decreases with increasing wavelength, which means that longer wavelengths of light are refracted less than shorter wavelengths. This difference in refraction causes the separation of colors in a prism or lens.

Resolving Power

Resolving power is a measure of the ability of an optical instrument to distinguish between two closely spaced objects. It is defined as the minimum distance between two points that can be distinguished as separate entities by the instrument.

In the context of wave optics, the resolving power of an instrument is limited by the phenomenon of diffraction. When light passes through an aperture or around an obstacle, it spreads out and interferes with itself, creating a diffraction pattern. This can cause closely spaced objects to appear blurred and indistinguishable.

The Rayleigh criterion is commonly used to determine the resolving power of an optical instrument. According to this criterion, two points are considered to be just resolved when the central maximum of the diffraction pattern of one point coincides with the first minimum of the diffraction pattern of the other point.

The resolving power of an instrument can be improved by increasing the size of the aperture or by using light of a shorter wavelength. This is because the angle of diffraction is inversely proportional to the size of the aperture and the wavelength of the light.

In summary, the resolving power of an optical instrument is its ability to distinguish between closely spaced objects, and is limited by the phenomenon of diffraction. The Rayleigh criterion is commonly used to determine the resolving power, and it can be improved by increasing the size of the aperture or using light of a shorter wavelength.

Rayleigh's criterion of resolution

Rayleigh's criterion of resolution is a standard for the minimum separation between two light sources that can be resolved by an optical instrument. It is based on the diffraction pattern produced by the instrument's aperture.

- According to Rayleigh's criterion, two point sources are considered to be just resolved when the central maximum of the diffraction pattern of one source coincides with the first minimum of the diffraction pattern of the other source.

- The minimum angular separation between two point sources that can be resolved by an optical instrument is given by the formula $\theta = 1.22 \lambda / D$, where λ is the wavelength of the light and D is the diameter of the instrument's aperture.
- Rayleigh's criterion is widely used in optical microscopy, astronomy, and other fields where high-resolution imaging is required.
- It is important to note that the criterion is a guideline and not an absolute limit. In some cases, it may be possible to resolve sources that are closer together than the criterion would suggest, while in other cases, sources that are farther apart may not be resolvable.

Resolving Power of Grating

The resolving power of a grating is defined as its ability to separate two closely spaced spectral lines. It is given by the formula:

$$R = nN$$

Where R is the resolving power, n is the order of the spectrum, and N is the total number of lines on the grating.

The resolving power of a grating increases with the order of the spectrum and the number of lines on the grating. A grating with a large number of lines and used in a high order of the spectrum can resolve very closely spaced spectral lines.

The resolving power of a grating can also be expressed in terms of the wavelength difference between two spectral lines that can be just resolved. This is given by the formula:

$$\Delta \lambda = \lambda / R$$

Where $\Delta \lambda$ is the wavelength difference between two spectral lines that can be just resolved, λ is the average wavelength of the two spectral lines, and R is the resolving power of the grating.

In summary, the resolving power of a grating is its ability to separate closely spaced spectral lines. It is determined by the order of the spectrum and the number of lines on the grating. The resolving power can also be expressed in terms of the wavelength difference between two spectral lines that can be just resolved. This is an important concept in the study of wave optics and is covered in Unit 3 of the subject of Engineering Physics.

Unit 4 - Fiber Optics & Laser

Fiber optics and lasers are two important technologies that have revolutionized the field of telecommunications and medicine. Here are some key points to know about these technologies:

1. **Fiber Optics** is a technology that uses thin, flexible fibers of glass or plastic to transmit light signals over long distances. The light signals carry information, such as voice, video, and data.
2. **Lasers** are devices that produce intense beams of light. They work by stimulating the emission of light from atoms or molecules. Lasers have many applications, including telecommunications, medicine, and manufacturing.
3. In telecommunications, fiber optics and lasers are used together to transmit information over long distances. Lasers generate the light signals, which are then sent through the fiber optic cables.
4. Fiber optic cables have several advantages over traditional copper cables. They can transmit information over longer distances, have higher bandwidth, and are immune to electromagnetic interference.
5. In medicine, lasers are used for a variety of procedures, including surgery, vision correction, and cancer treatment. They offer precision and control, minimizing damage to surrounding tissue.
6. Fiber optics and lasers have also found applications in other fields, such as manufacturing, where they are used for cutting, welding, and marking materials.
7. The development of fiber optics and lasers has led to many advancements and has greatly impacted our daily lives. They continue to be an active area of research and development, with new applications being discovered regularly.

Fibre Optics: Principle and construction of optical fiber

Fiber optics is the science of transmitting data, voice, and images by the passage of light through thin, transparent fibers. In telecommunications, fiber optic technology has virtually replaced copper wire in long-distance telephone lines, and it is used to link computers within local area networks.

Principle

Optical fibers typically work on the principle of total internal reflection of light. The term total internal reflection indicates that no part of the signal gets refracted or transmitted to the second medium, but instead, all of the signals is retained inside the fiber.

Construction

Modern optical fibers are made of double layers of glass. The inner fiber core (8 μm) is encircled by cladding (125 μm), a concentric core of lower index glass. A protective layer surrounds the cladding. The core-cladding interface is where total internal reflection occurs.

The optical fiber is a device that works on the principle of total internal reflection by which light signals can be transmitted from one place to another with a negligible loss of energy. It has a large bandwidth. The optical frequency of 2×10^{14} Hz can be used and hence the system has higher bandwidth.

Acceptance Angle

- The acceptance angle of an optical fiber is defined based on a purely geometrical consideration (ray optics): it is the maximum angle of a ray (against the fiber axis) hitting the fiber core which allows the incident light to be guided by the core.
- The maximum incidence angle of a light ray which can be used for injecting light into a fiber core or waveguide is known as the Acceptance Angle.
- Acceptance angle is just the conical half angle of the acceptance cone.
- Wave optics is required for fully understanding the details of launching light into fibers.
- This parameter is commonly used in lieu of the acceptance angle in datasheets for fiber optic cable.

Numerical Aperture

Numerical Aperture (NA) is a measure of the ability of an optical fiber to collect or confine the incident light ray inside it. It is among the most basic property of optical fiber and shows the efficiency with which light is collected inside the fiber in order to get propagated.

In microscopy, numerical aperture is commonly used to describe the acceptance cone of an objective (and hence its light-gathering ability and resolution). In fiber optics, it describes the range of angles within which light that is incident on the fiber will be transmitted along it.

The numerical aperture of an optical system (e.g. an imaging system) is a measure for its angular acceptance for incoming light. It is defined based on geometrical considerations and is thus a theoretical parameter which is calculated from the optical design.

In the context of fiber optics, the numerical aperture can be found using the equation (4). It is also related to the acceptance angle and acceptance cone of the optical fiber.

Acceptance Cone

- The acceptance cone is the cone within which optical power may be coupled into the bound modes of an optical fiber .
- The cross section of an optical fiber is circular; the light waves accepted by the core are expressed as a cone .
- The larger the acceptance cone, the larger the numerical aperture (NA); this means that the fiber is able to accept and propagate more light .
- The acceptance cone is derived by rotating the acceptance angle about the fiber axis .
- In order to get high light transmission, light should propagate through the fiber by a series of total internal reflections. This transmission is achieved if the angles of the light entering the fiber are within the acceptance cone .

Step Index and Graded Index Fibers

Introduction

Step index and graded index fibers are two types of optical fibers used in fiber optics. They differ in the way the refractive index of the core changes from the center to the cladding.

Step Index Fiber

- A step index fiber has a core with a constant refractive index and a sharp decrease in the index of refraction at the cladding .
- Step index fiber is often used for short distance communication .
- Step index fibers are less expensive to manufacture than graded index fibers .
- They have a higher numerical aperture, which allows them to collect light from a wider range of angles .
- Step index fibers have higher dispersion .

Graded Index Fiber

- A graded index fiber has a core where the refractive index is maximum at the center and decreases gradually towards the core-cladding interface .
- Graded index fiber is only of one type, that is multi mode fiber .
- In graded index fiber, the refractive index of the core gradually lowers towards the cladding, which causes light to travel faster in lower refractive index .

Conclusion

In summary, step index and graded index fibers are two types of optical fibers used in fiber optics. They differ in the way the refractive index of the core changes from the center to the cladding. Step index fibers are often used for short distance communication and are less expensive to manufacture than graded index fibers. Graded index fibers have a core where the refractive index decreases gradually towards the core-cladding interface.

Fiber Optic Communication Principle

Fiber optic communication is a method of transmitting information from one place to another by sending pulses of infrared light through an optical fiber. The light is a form of carrier wave that is modulated to carry information .

The fiber optic works on the principle of total internal reflection. When a Ray of light is incident on the core of the optical fiber at a small angle, it suffers refraction and strikes the core cladding interface. As the diameter of the fiber is very small hence the angle of incidence is greater than the critical angle .

Optical fiber is used by telecommunications companies to transmit telephone signals, Internet communication and cable television signals. It is also used in other industries, including medical, defense, government, industrial and commercial .

First, an electrical signal is converted into an optical signal, and then an optical signal is transmitted through the optical fiber, which is a type of wired communication .

In telecommunications, fiber optic technology has virtually replaced copper wire in long-distance telephone lines, and it is used to link computers within local area networks .

Attenuation in Fiber Optics

Attenuation in fiber optics, also known as transmission loss, is the reduction in intensity of the light beam (or signal) with respect to distance traveled through a transmission medium.

- Attenuation can occur to any kind of signal like fiber, copper, satellite, fiber, etc. In the Fiber signal, it travels on HF (high-frequency) wavelength light which can be protected by glass tubes.
- The attenuation coefficient of FOC (fiber optic cable) is one of the most significant parameters. In a huge amount, the distance of relay can be decided within the optical transmission. The attenuation coefficient of fiber can be 0.36dB/km in wavelength of 1310nm as well as 0.22dB/km in wavelength of 1550nm.

- Attenuation, in any form, is the loss of power (or flux intensity) over a long distance when the signal in question is transmitted through a medium. In fiber optics, this is the loss of light intensity through the fiber optic between the source and the receiver.
- Attenuation (dB) = $10 \times \log_{10} (P_{\text{in}}/P_{\text{out}}) = 20 \times \log_{10} (V_{\text{in}}/V_{\text{out}})$
Gain (dB) = $10 \times \log_{10} (P_{\text{out}}/P_{\text{in}}) = 20 \times \log_{10} (V_{\text{out}}/V_{\text{in}})$.
- Optical Fiber Structure Optical fiber is a medium to carry information. Optical fiber is made of silica-based glass, and consists of a core surrounded by cladding.

Dispersion in Fiber Optics

Dispersion is a phenomenon that occurs when light travels through a medium, such as an optical fiber. It is the spreading of light pulses as they travel along the fiber, which can cause distortion of the signal being transmitted. There are several types of dispersion that can occur in optical fibers, including:

1. **Chromatic Dispersion:** This type of dispersion occurs due to the different wavelengths of light traveling at different speeds through the fiber. This causes the light pulses to spread out as they travel along the fiber, leading to signal distortion.
2. **Modal Dispersion:** This type of dispersion occurs in multimode fibers, where different modes of light travel at different speeds through the fiber. This causes the light pulses to spread out as they travel along the fiber, leading to signal distortion.
3. **Polarization Mode Dispersion:** This type of dispersion occurs due to the different polarization states of light traveling at different speeds through the fiber. This causes the light pulses to spread out as they travel along the fiber, leading to signal distortion.

To minimize the effects of dispersion, optical fibers are designed with specific properties, such as the refractive index profile, to reduce the spreading of light pulses as they travel along the fiber. Additionally, dispersion compensating fibers and other techniques can be used to mitigate the effects of dispersion in optical communication systems.

Unit 4 - Fiber Optics & Laser: Application of Fiber

Fiber optics is a technology that uses glass or plastic fibers to transmit data. It has many applications in various fields, including:

1. **Telecommunications:** Fiber optic cables are used to transmit telephone, internet, and cable television signals over long distances. They offer higher

bandwidth and faster data transmission rates than traditional copper cables.

2. **Medical:** Fiber optics is used in medical equipment such as endoscopes, which allow doctors to see inside the body without performing surgery. They are also used in surgical instruments, such as lasers, to perform minimally invasive procedures.
3. **Sensors:** Fiber optic sensors are used to measure temperature, pressure, and other physical properties. They are used in various industries, including oil and gas, aerospace, and civil engineering.
4. **Lighting:** Fiber optic lighting is used in decorative and architectural applications, such as in museums, theaters, and swimming pools. It is also used in automotive and aircraft lighting.
5. **Military:** Fiber optics is used in military communications and surveillance systems. It offers secure and reliable data transmission, even in harsh environments.

These are just a few examples of the many applications of fiber optics. The technology continues to evolve, and new applications are being developed all the time.

Laser: Absorption of radiation

- Absorption of radiation is the process by which electrons in the ground state absorb energy from photons to jump into a higher energy level.
- When laser radiation is applied to the surface of a material for a certain time, part of it is absorbed by the conduction electrons in a surface region a few nanometers thick and rapidly converted into heat through collisions between the electrons and the lattice ions .
- Lasers can be absorbed inside an optical substrate through several different methods. Electrons in discrete energy levels of the atoms that make up the optical medium absorb radiative photons and are pushed to semi-stable, higher energy levels.
- Absorption is the process in which optical energy is converted to internal energy of electrons, atoms, or molecules. When a photon is absorbed, the energy may cause an electron in an atom to go from a lower to a higher energy level, thereby changing the internal momentum of the electron and the electron's internal quantum numbers.

Spontaneous and Stimulated Emission of Radiation

Spontaneous emission is the process by which an excited atom or molecule, without any external influence, releases energy in the form of a photon and

returns to a lower energy state. This process is random and the direction, phase, and polarization of the emitted photon are unpredictable.

Stimulated emission, on the other hand, occurs when an incoming photon, with energy equal to the energy difference between the excited state and a lower energy state of an atom or molecule, interacts with the excited atom or molecule and causes it to release a second photon. This second photon is identical to the incoming photon in terms of direction, phase, and polarization.

The process of stimulated emission is the basis for the operation of lasers. In a laser, a population inversion is created, where the number of atoms or molecules in an excited state is greater than the number in a lower energy state. This allows for a chain reaction of stimulated emission, where each emitted photon stimulates the emission of another photon, leading to the amplification of light.

In summary: - Spontaneous emission is the random release of energy in the form of a photon by an excited atom or molecule, without any external influence. - Stimulated emission occurs when an incoming photon interacts with an excited atom or molecule and causes it to release a second, identical photon. - The process of stimulated emission is the basis for the operation of lasers, where a population inversion is created to allow for the amplification of light.

This concludes the notes on Spontaneous and Stimulated Emission of Radiation for Unit 4 - Fiber Optics & Laser in the subject of ENGINEERING PHYSICS.

Population Inversion

- Population inversion is a technique mainly used for light amplification .
- It is required for laser operation .
- Consider a group of electrons with two energy levels E_1 and E_2 .
- A population inversion ($N_2 > N_1$) has thus been achieved between level 1 and 2, and optical amplification at the frequency ω_1 can be obtained .
- At least half the population of atoms must be excited from the ground state to obtain a population inversion, the laser medium must be very strongly pumped .
- Population inversion, and thus optical gain in the sense of Eq. 14, may be obtained in a direct bandgap semiconductor exhibiting radiative recombination of electrons and holes provided there is, simultaneously in space and time, a large density of electrons in the conduction band and holes in the valence band such as that provided by the junction region of a diode under heavy forward bias where large numbers of electrons and holes exist .
- Population inversion and laser action have been discovered in a large number of optical transitions in gases and gas mixtures .
- A gas as the laser active medium has several advantages: scalability to large volume and effective cooling by diffusion or by convective flow, allowing potentially high power output .

- Normally, a system of atoms is in temperature equilibrium and there are always more atoms in low energy states than in higher ones .

Einstein's Coefficients

Einstein's coefficients are a set of coefficients that relate the rates of absorption, spontaneous emission, and stimulated emission of electromagnetic radiation in a system of particles. These coefficients were introduced by Albert Einstein in 1917 to explain the interaction of radiation with matter.

1. **Absorption:** The absorption coefficient, denoted by B_{12} , represents the probability of a particle in the lower energy state 1 absorbing a photon and transitioning to the higher energy state 2.
2. **Spontaneous emission:** The spontaneous emission coefficient, denoted by A_{21} , represents the probability of a particle in the higher energy state 2 spontaneously emitting a photon and transitioning to the lower energy state 1.
3. **Stimulated emission:** The stimulated emission coefficient, denoted by B_{21} , represents the probability of a particle in the higher energy state 2 being stimulated by a photon to emit another photon and transition to the lower energy state 1.

Einstein derived the relationship between these coefficients, showing that $B_{12} = B_{21}$ and that the ratio of A_{21} to B_{21} is equal to the ratio of the energy density of the radiation field to the energy of a single photon.

These coefficients play a crucial role in the operation of lasers, where the population inversion between two energy levels is achieved by pumping particles into the higher energy state, leading to a high rate of stimulated emission and the generation of coherent light. They are also important in the study of absorption and emission spectra of materials, providing a quantitative description of the interaction of radiation with matter.

Principles of Laser Action

Laser action is based on the principles of stimulated emission and population inversion. Here are the key points to understand these principles:

1. **Stimulated Emission:** When an atom is in an excited state, it can release its excess energy in the form of a photon. This process is called spontaneous emission. However, if an external photon with the same energy as the energy difference between the excited state and the ground state interacts with the excited atom, it can stimulate the atom to release its excess energy in the form of a second photon. This process is called stimulated emission. The two photons are identical in energy, phase, and direction of travel.

2. **Population Inversion:** In order for stimulated emission to dominate over absorption, the number of atoms in the excited state must be greater than the number of atoms in the ground state. This condition is called population inversion. Population inversion can be achieved by pumping energy into the system, either optically or electrically.
3. **Optical Resonator:** A laser consists of an optical resonator, which is a cavity with mirrors at both ends. The mirrors reflect the photons back and forth, allowing them to interact with the atoms in the gain medium multiple times. This increases the probability of stimulated emission and amplifies the light.
4. **Lasing Threshold:** When the gain due to stimulated emission exceeds the losses due to absorption and mirror transmission, the system reaches the lasing threshold and laser action begins. The output of the laser is a coherent beam of light with a narrow spectral linewidth.

These are the basic principles of laser action. They are essential for understanding the operation of a laser and are covered in Unit 4 - Fiber Optics & Laser in the subject of ENGINEERING PHYSICS.

Unit 4 - Fiber Optics & Laser in the subject of ENGINEERING PHYSICS

Solid state Laser (Ruby laser) and Gas Laser (He-Ne laser)

- **Solid state lasers** use a solid material as the active medium, where the laser light is generated. One example of a solid state laser is the **Ruby laser**.
- The Ruby laser uses a synthetic ruby crystal as the active medium, which is pumped with a high-intensity light source to produce laser light.
- The ruby crystal is made of aluminum oxide, with some of the aluminum ions replaced by chromium ions. The chromium ions absorb the pump light and emit light at a wavelength of 694.3 nm, in the red part of the spectrum.
- Ruby lasers are typically pulsed, meaning they emit light in short bursts, rather than continuously.
- **Gas lasers** use a gas as the active medium. One example of a gas laser is the **Helium-Neon (He-Ne) laser**.
- The He-Ne laser uses a mixture of helium and neon gases as the active medium, contained in a glass tube. The gas mixture is excited by an electric discharge, causing the neon atoms to emit light at a wavelength of 632.8 nm, in the red part of the spectrum.
- He-Ne lasers are typically continuous-wave, meaning they emit light continuously, rather than in pulses.
- Both Ruby and He-Ne lasers are commonly used in scientific research, as

well as in various industrial and medical applications.

Laser Applications

Lasers have a wide range of applications in various fields. Here are some of the most common applications of lasers in the field of engineering physics:

1. **Communication:** Lasers are used in fiber optic communication systems to transmit data over long distances. The laser light is modulated to carry the information and is sent through an optical fiber.
2. **Medicine:** Lasers are used in various medical procedures such as laser eye surgery, laser hair removal, and laser tattoo removal. They are also used in medical imaging and diagnostics.
3. **Industry:** Lasers are used in various industrial processes such as cutting, welding, and drilling. They are also used for marking and engraving.
4. **Measurement:** Lasers are used for precise measurements in various fields such as surveying, construction, and manufacturing. They are also used in scientific research for measuring distances, velocities, and other physical quantities.
5. **Entertainment:** Lasers are used in various forms of entertainment such as laser light shows, laser tag, and laser projection.

These are just a few of the many applications of lasers in the field of engineering physics. Lasers have revolutionized many fields and continue to be an important tool in research and industry.

Unit 5 - Superconductors and Nano-Materials:

Superconductors are materials that exhibit zero electrical resistance and the expulsion of magnetic fields below a certain temperature, known as the critical temperature. This phenomenon is known as superconductivity.

Nano-materials are materials with at least one dimension in the nanometer range (1-100 nm). These materials exhibit unique properties due to their small size and high surface area to volume ratio.

1. **Superconductors:**
 - Superconductivity was first discovered in 1911 by Heike Kamerlingh Onnes.
 - The critical temperature of a superconductor is the temperature below which the material exhibits zero electrical resistance.
 - The Meissner effect is the expulsion of magnetic fields from a superconductor below its critical temperature.
 - Type I superconductors are materials that exhibit superconductivity below a certain critical temperature and magnetic field strength.

- Type II superconductors are materials that exhibit superconductivity below a certain critical temperature and can withstand higher magnetic field strengths than Type I superconductors.
- High-temperature superconductors are materials that exhibit superconductivity at temperatures above the boiling point of liquid nitrogen (-196°C).
- Applications of superconductors include MRI machines, particle accelerators, and power transmission.

2. Nano-Materials:

- Nano-materials can be classified into zero-dimensional (e.g. nanoparticles), one-dimensional (e.g. nanowires), two-dimensional (e.g. graphene), and three-dimensional (e.g. nanostructured materials).
- The properties of nano-materials can be different from their bulk counterparts due to their small size and high surface area to volume ratio.
- Nano-materials can exhibit unique optical, electronic, magnetic, and mechanical properties.
- Applications of nano-materials include drug delivery, electronics, energy storage, and catalysis.

Superconductors: Temperature dependence of resistivity in superconducting materials

- Superconductors are special materials in which after a certain critical temperature the resistance drops to zero value and the conductivity thus reaches the maximum .
- Superconductivity is a phenomena in certain metals and ceramics where the resistivity of the material drops to zero below a certain critical temperature known as TC .
- These temperatures are low, with most industrial superconductors below 12 K and higher temperature ceramics just above 130K .
- The resistivity of materials depend on the temperature. $\rho = \rho_0 [1 + \alpha(T - T_0)]$ is the equation that shows the relation between the temperature and the resistivity of a material .
- In the equation ρ_0 is the resistivity at a standard temperature, ρ is the resistivity at T °C, T_0 is the reference temperature and α is the temperature coefficient of resistance .

Meissner Effect

The Meissner effect is a property of all superconductors. It was discovered by the German physicists W. Meissner and R. Ochsenfeld in 1933. The Meissner effect is the expulsion of a magnetic field from the interior of a material that is in the

process of becoming a superconductor. This means that when a superconductor is cooled below its transition temperature, usually close to absolute zero, it loses its resistance to the flow of electrical currents and all or part of the magnetic field within the material is expelled.

One of the most well-known demonstrations of the Meissner effect is its ability to make a magnet levitate above a superconductor. This effect of superconductivity is used in magnetic levitation, which is the base of modern high-speed bullet trains.

In summary, the Meissner effect is a property of superconductors where a magnetic field is expelled from the interior of the material when it is cooled below its transition temperature. This effect has practical applications, such as in magnetic levitation.

Temperature dependence of critical field

The critical field of a superconductor is the magnetic field strength at which the superconductor transitions from the superconducting state to the normal state. The critical field is temperature dependent and decreases as the temperature of the superconductor increases.

The relationship between the critical field and temperature can be described by the equation:

$$H_c(T) = H_c(0)(1 - (T/T_c)^2)$$

where $H_c(T)$ is the critical field at temperature T , $H_c(0)$ is the critical field at absolute zero, and T_c is the critical temperature of the superconductor.

This equation shows that as the temperature of the superconductor increases, the critical field decreases. At the critical temperature, the critical field becomes zero and the superconductor transitions to the normal state.

This temperature dependence of the critical field is an important factor to consider when designing and using superconducting materials in applications such as MRI machines and particle accelerators. It is important to ensure that the operating temperature and magnetic field strength are within the limits of the superconductor's critical field to maintain its superconducting properties.

Persistent Current

- A persistent current is a steady electric current that flows indefinitely in a superconducting loop.
- This phenomenon occurs because superconductors have zero electrical resistance, allowing the current to flow without any energy loss.
- The current will continue to flow until an external force, such as a magnetic field or heat, disrupts the superconducting state.

- Persistent currents are used in a variety of applications, including superconducting magnets and energy storage devices.
- In superconducting magnets, the persistent current generates a strong magnetic field that can be used for medical imaging, particle accelerators, and other applications.
- In energy storage devices, the persistent current can be used to store electrical energy with very high efficiency.

Type I and Type II Superconductors

Type I Superconductors

- Type I superconductors have only one critical magnetic field. When that value is reached, the superconductive characteristics of a material are gone.
- Transition from normal state to superconducting state occurs instantly.
- Magnetic field lines cannot penetrate through superconductors.
- Type-I superconductors are generally pure metals.

Type II Superconductors

- Type II superconductors, on the other hand, have two critical magnetic fields.
- Transition happens slowly i.e. by decreasing temperature after critical temperature superconducting properties increase.
- Some magnetic field lines can penetrate through superconductor.
- Most type II superconductors have higher critical temperature.
- Type-II superconductors are usually made of metal alloys or complex oxide ceramics.
- All high temperature superconductors are type-II superconductors.
- While most elemental superconductors are type-I, niobium, vanadium, and technetium are elemental type-II superconductors.
- Boron-doped diamond and silicon are also type-II superconductors.
- Type II superconductors lose their superconductivity gradually but not easily or abruptly when placed in the external magnetic field.

High Temperature Superconductors

- High-temperature superconductors (abbreviated high-T_c or HTS) are defined as materials that behave as superconductors at temperatures above 77 K (−196.2 °C; −321.1 °F), the boiling point of liquid nitrogen.
- The main class of high-temperature superconductors is copper oxides combined with other metals, especially the Rare-earth barium copper oxides (REBCOs) such as Yttrium barium copper oxide (YBCO).
- The second class of high-temperature superconductors in the practical classification is the iron-based compounds.

- High-temperature superconductors (HTSCs), when discovered in 1986, presented a huge challenge but also opportunity to companies like ABB, GE, and Siemens to radically improve their products.
- A new experiment led by the condensed matter physicist Séamus Davis at the University of Oxford all but settles the origin of high-temperature superconductivity, a puzzle Davis has worked on for 25 years. When electrons couple up, further quantum trickery makes superconductivity unavoidable.

Properties and Applications of Super-conductors

Properties of Super-conductors

- A superconductor is an element or metallic alloy which, when cooled below a certain threshold temperature, the material dramatically loses all electrical resistance.
- In principle, superconductors can allow electrical current to flow without any energy loss (although, in practice, an ideal superconductor is very hard to produce).
- Superconductors are also able to maintain a current with no applied voltage whatsoever.
- Superconductivity occurs for magnetic fields and temperatures below the curves shown.
- Another important property of a superconducting material is its critical magnetic field $B_c(T)$, which is the maximum applied magnetic field at a temperature T that will allow a material to remain superconducting.

Applications of Super-conductors

- Superconductors are used in particle accelerators, generators, transportation, computing, electric motors, medical, power transmission, etc.
- Superconductors are primarily employed for creating powerful electromagnets in MRI scanners.
- These conductors are used to transmit power for long distances.
- Most of the applications of the superconductor examples are because of their properties which provide advantages such as low power loss because of less dissipation of energy, high-speed operations because of zero resistance and continuous flowing electrical current, and high sensitivity.
- Experiments have demonstrated that currents in superconducting coils can persist for years without any measurable degradation.

Unit 5 - Superconductors and Nano-Materials

Nano-Materials: Introduction and properties of nano materials

Nano-materials are materials that have at least one dimension in the nanometer scale, typically between 1 and 100 nanometers. At this scale, materials exhibit unique properties that are not observed in their bulk form. These properties are due to the increased surface area to volume ratio and quantum confinement effects.

Some of the unique properties of nano-materials include:

1. Increased reactivity: Due to the increased surface area, nano-materials are more reactive than their bulk counterparts. This property is useful in applications such as catalysis and drug delivery.
2. Enhanced mechanical properties: Nano-materials can have enhanced mechanical properties such as increased strength and toughness. This is due to the presence of a large number of grain boundaries and the ability to control the microstructure at the nanoscale.
3. Optical properties: Nano-materials can exhibit unique optical properties such as fluorescence and surface plasmon resonance. These properties are useful in applications such as sensing and imaging.
4. Electrical properties: Nano-materials can exhibit unique electrical properties such as increased conductivity and quantum tunneling. These properties are useful in applications such as electronics and energy storage.
5. Magnetic properties: Nano-materials can exhibit unique magnetic properties such as superparamagnetism and giant magnetoresistance. These properties are useful in applications such as data storage and spintronics.

In summary, nano-materials are materials with at least one dimension in the nanometer scale and exhibit unique properties due to their increased surface area to volume ratio and quantum confinement effects. These properties include increased reactivity, enhanced mechanical properties, unique optical, electrical, and magnetic properties. These properties make nano-materials useful in a wide range of applications.

Unit 5 - Superconductors and Nano-Materials: Basics concept of Quantum Dots

- Quantum dots are colloidal fluorescent semiconductor nanocrystals, roughly spherical in shape .
- They typically have unique optical, electronic and photophysical properties that make them appealing in promising applications in biological

labeling, imaging, and detection and as efficient fluorescence resonance energy transfer donors .

- Quantum dots are also called semiconductor nanocrystals, and are semiconductor particles a few nanometres in size .
- Their optical and electronic properties differ from those of larger particles as a result of quantum mechanics .
- Quantum dots defined by lithographically patterned gate electrodes, or by etching on two-dimensional electron gases in semiconductor heterostructures can have lateral dimensions between 20 and 100 nm .
- Some quantum dots are small regions of one material buried in another with a larger band gap .
- Quantum dots are also brighter than a rival technology known as organic LEDs (OLEDs) and could potentially make OLED displays obsolete .
- Quantum dots can be used to make the red, green, and blue pixels in TV screens with brighter and more precise colors than in traditional LCDs or rival technologies such as OLEDs .

Quantum Wires and Quantum Wells

Quantum wires and quantum wells are semiconductor structures that confine the motion of electrons in one or two dimensions, respectively. These structures have applications in nanoelectronics, optoelectronics, information technology, and biotechnology, due to the quantum mechanical confinement provided by the small spatial extent of the structures.

Quantum Wells

Quantum well materials are made of thin layers of differing semiconductor materials, and can be grown such that the movement of the free electron is confined to a two-dimensional plane. This is known as a two-dimensional (2D) structure or quantum well, where quantization of the particle motion occurs in one direction, while the particle is free to move in the other two directions.

Quantum Wires

Quantum wires, on the other hand, are one-dimensional (1D) structures where quantization occurs in two directions, leading to free movement along only one direction. For metals, quantization corresponding to the lowest energy states is only observed for atomic wires. Their corresponding wavelength being thus extremely small they have a very large energy separation which makes resistance quantization observable even at room temperature.

Fabrication of Nano Materials

There are two main approaches to the fabrication of nano materials: the top-down approach and the bottom-up approach.

Top-Down Approach (CVD)

The top-down approach involves breaking down larger materials into smaller nano-sized particles. One common method used in this approach is Chemical Vapor Deposition (CVD). In CVD, a gas-phase precursor is introduced into a reaction chamber, where it decomposes and deposits onto a substrate to form a thin film of the desired material. The properties of the film can be controlled by adjusting the deposition parameters, such as temperature, pressure, and precursor flow rate.

Bottom-Up Approach (Sol Gel)

The bottom-up approach involves building up nano-sized particles from smaller building blocks, such as atoms or molecules. One common method used in this approach is the Sol Gel process. In the Sol Gel process, a solution of precursor molecules is prepared and then allowed to undergo a controlled chemical reaction to form a gel. The gel is then dried and heat-treated to form a solid material with nano-sized particles. The properties of the material can be controlled by adjusting the reaction conditions, such as temperature, pH, and precursor concentration.

These are the two main approaches to the fabrication of nano materials, each with its own advantages and disadvantages. The choice of approach depends on the specific requirements of the application and the properties of the desired material.

Properties and Application of Nano Materials

Nanomaterials are materials with a particle size of less than 100 nm in at least one of its dimensions . They possess unique optical, magnetic, electrical, and other properties that emerge at this scale . These emergent properties have great potential applications in various fields such as electronics, medicine, and environmental improvement .

Some of the properties and applications of nanomaterials are:

1. **Optical properties:** The optical properties of nanomaterials are used to form optical detectors, sensors, lasers, displays, and solar cells. This property is also used in biomedicine and photoelectrochemistry .
2. **Electrical properties:** Nanomaterials have unique electrical properties that can be exploited in the field of electronics. For example, in microbial fuel cells, the electrodes are made up of carbon nanotubes .

3. **Magnetic properties:** The magnetic properties of nanomaterials have potential applications in the field of medicine, such as in drug delivery and magnetic resonance imaging .
4. **Environmental remediation:** Nanomaterials have potential applications in environmental remediation, such as in the removal of pollutants from water and air .

These are just a few examples of the properties and applications of nanomaterials. The field of nanotechnology is constantly evolving, and new applications are being discovered all the time.